

Explainable Audio Deepfake Detection Using wav2vec 2.0 Embeddings and Multilayer Perceptron Classification

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Abstract—The rapid development of neural text-to-speech, voice conversion, and voice cloning systems has increased the risk of synthetic speech being misused for impersonation, fraud, misinformation, and evidence manipulation. This paper presents an explainable audio deepfake detection framework using mean-pooled wav2vec 2.0 BASE embeddings and a multilayer perceptron (MLP) classifier. The Fake-or-Real (FoR) dataset is used as the experimental corpus, including the clean for-norm condition and the loudspeaker-microphone re-recorded for-rerec condition. Audio signals are resampled to 16 kHz, segmented into fixed 4-second clips, represented using wav2vec 2.0 Layer 4 embeddings, and converted into 768-dimensional vectors through temporal mean pooling. Class imbalance is handled by applying SMOTE to the training set only, while validation and test distributions are kept unchanged. Architecture variation, threshold calibration, classifier comparison, and explainability analysis are conducted to evaluate the proposed framework. The selected MLP architecture [512, 384, 256] with threshold 0.3 achieves 93.28% accuracy, 98.14% precision, 85.55% recall, 91.41% F1-score, and 0.9830 AUC on the pooled FoR test set. Compared with Random Forest and XGBoost baselines, XGBoost obtains the highest raw F1-score, but the MLP is selected because it provides the highest precision and supports gradient-based SHAP explanation. SHAP GradientExplainer identifies feat 153 as the most influential embedding dimension. Cumulative SHAP analysis shows that K=200 features retain 55.1% of cumulative importance, K=374 features retain 80.0%, and K=572 features retain 95.0%. Preliminary pooled-test error inspection suggests a potential false-negative risk under channel-degraded audio, but separate for-rerec subset evaluation is left for future work.

Index Terms—audio deepfake detection, wav2vec 2.0, multilayer perceptron, SHAP, explainable AI, feature importance, Fake-or-Real dataset, cross-condition robustness

I. INTRODUCTION

Audio deepfakes are artificially synthesized or converted speech signals designed to imitate human speakers. They are commonly generated using neural text-to-speech (TTS), voice conversion (VC), neural vocoders, or short-sample voice cloning systems. While these technologies support useful applications such as accessibility tools and speech restoration, they also create risks for voice-based authentication, digital forensics, financial fraud, misinformation, and legal evidence manipulation [1]–[3].

The urgency of audio deepfake detection has increased because synthetic speech generation is becoming more accessible and realistic. Recent surveys report that detection research has shifted from simple in-domain classification toward more difficult requirements, including cross-dataset generalization, robustness to acoustic degradation, and explainability of model decisions [1], [4], [5]. These requirements are especially important because a detector that performs well on clean recordings may fail when audio is compressed, transmitted through a

communication channel, or re-recorded through a loudspeaker-microphone setup [6]–[8].

Traditional spoofing countermeasures have often relied on handcrafted acoustic features or spectrogram-based convolutional models. Recent work shows that self-supervised speech representations are promising for audio deepfake detection because they encode phonetic, spectral, and contextual cues learned from large-scale unlabeled audio [9]–[11]. In particular, wav2vec 2.0 has been applied successfully to automatic speaker verification anti-spoofing and speech deepfake detection, motivating its use as the feature extractor in this study [9].

Despite these advances, several issues remain underexplored. First, the optimal depth and width of MLP classifiers on dense wav2vec embedding spaces is rarely analyzed systematically. Second, threshold calibration under class imbalance is often less emphasized than model architecture. Third, although explainable AI is increasingly recognized as important for trustworthy deepfake detection, gradient-based SHAP analysis on dense self-supervised speech embeddings remains limited. Fourth, the minimum subset of embedding dimensions required for reliable detection is not well established. Fifth, robustness under acoustic channel degradation requires further investigation, especially when clean and re-recorded conditions are mixed. Sixth, despite having been used in anti-spoofing and speech deepfake detection, reviewed implementation applied fine-tuning on the wav2vec 2.0 model. A certain limitation on the fine-tuning method is the dataset-specific biased output. Therefore, this study explores the characteristics of deepfake audio itself.

This study addresses these issues using the FoR dataset. The main contributions are as follows: (1) wav2vec 2.0 layerwise ablation study; (2) evaluation of MLP architecture variations on the best wav2vec 2.0 ablated layer; (3) threshold calibration across 0.3, 0.4, 0.5, and 0.6; (4) classifier comparison against Random Forest and XGBoost baselines; (5) SHAP GradientExplainer analysis for global feature importance; (5) cumulative SHAP-based feature subset analysis showing that K=374 retains 80.0% of total importance, while K=200 provides a compact 55.1% subset; and (6) preliminary cross-condition discussion based on pooled-test error inspection.

II. RELATED WORK

A. Audio Deepfake Detection and Benchmarks

Recent audio deepfake detection research has expanded from binary synthetic speech classification to broader tasks such as partially fake audio localization, algorithm attribution, robustness evaluation, and in-the-wild detection [12]–[14]. The ADD 2022 and ADD 2023 challenges introduced more realistic evaluation settings, including partially fake audio and multiple deepfake analysis subtasks [12], [13]. Recent surveys further emphasize that dataset diversity and acoustic variability remain central challenges for the field [1], [4], [5].

Cross-domain and multilingual benchmarks have also become important because detectors trained on one corpus may not generalize

to new languages, speakers, vocoders, codecs, or recording channels. Cross-domain datasets and studies show that robust detection requires evaluation beyond a single clean condition [6], [15], [16]. This motivates the use of both for-norm and for-rerec conditions in the present study.

B. Self-Supervised Speech Representations

Self-supervised speech models have become strong feature extractors for synthetic speech detection. Tak et al. demonstrated the effectiveness of wav2vec 2.0 with augmentation for spoofing and deepfake detection [9]. Kawa et al. showed that Whisper features are also useful for deepfake detection [10]. Martin-Donas et al. explored self-supervised embeddings and synthetic data augmentation for robust audio deepfake detection, showing the value of pretrained representations as front-end feature extractors [11]. Other recent work has also investigated sensitive layer selection and SSL-based front-end robustness [17], [18].

C. Explainability and Feature Importance

Explainability is increasingly important because high accuracy alone does not explain why a detector labels audio as fake. Recent studies have applied explainable methods to deepfake audio detection using SHAP, LIME, Grad-CAM, and related attribution approaches [19]–[21]. Explainability is particularly relevant for self-supervised embeddings because their dimensions are latent and not directly interpretable as conventional acoustic descriptors. Therefore, feature attribution can help identify which embedding dimensions contribute most strongly to classification.

D. Robustness and Cross-Condition Generalization

Generalization remains one of the major challenges in audio deepfake detection. Muller et al. showed that audio deepfake detectors may degrade when evaluated outside their training domain [6]. Calibration and generalizable synthetic speech detection have also been studied using pretrained self-supervised representations [7]. Recent works further highlight the importance of data-centric evaluation, codec and channel robustness, and in-the-wild audio deepfake detection [8], [14], [22]. These findings support the need to evaluate or at least inspect model behavior under re-recorded or channel-degraded conditions.

III. MATERIALS AND METHODS

A. Dataset

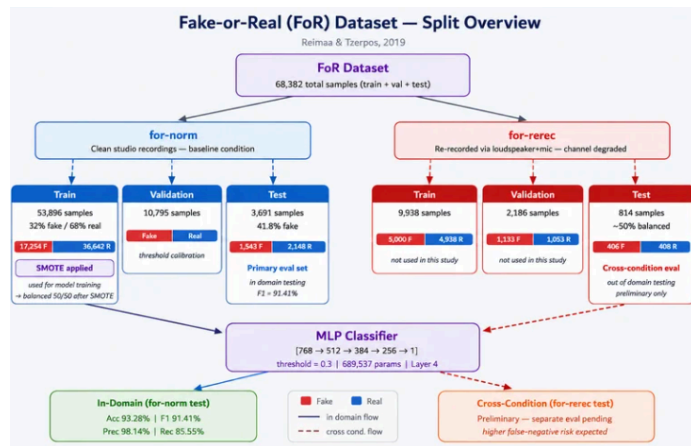


Fig. 1. Fake-or-Real dataset split overview, including clean for-norm and re-recorded for-rerec conditions.

This study uses the Fake-or-Real (FoR) dataset as the experimental corpus. Two conditions are considered: for-norm and for-rerec. The for-norm subset contains clean studio recordings and is used as the main clean baseline condition. The for-rerec subset contains recordings re-recorded through a loudspeaker-microphone channel, introducing room acoustics, frequency response distortion, and background noise.

Recent survey literature describes FoR as one of the established datasets used in synthetic speech detection research [1].

The experimental split used in this study contains 53,896 training samples, 10,795 validation samples, and 3,691 test samples. The training set contains 17,254 fake samples and 36,642 real samples, corresponding to 32.0% fake samples. The test set contains 1,543 fake samples and 2,148 real samples, corresponding to 41.8% fake samples. This distribution shift motivates both class imbalance handling and threshold calibration.

B. Preprocessing and Segmentation

All audio files are resampled to 16 kHz. Each audio signal is segmented into a fixed-length. Fixed-length segmentation standardizes the input duration for MLP. Duration study was conducted to find the best duration that preserves the most data. The audios are segmented with upper threshold and lower threshold. Upper threshold determines the fixed-length of the clips, and lower threshold determines the skipping boundary for the system. Lower threshold (set to either half or one-third of the upper threshold) is useful to prevent too much 0s padding on each clip.

The duration study shows an upper boundary of 4-second with low threshold of 1.33 (one-third of the upper boundary) provides the best performance. Preliminary raw duration inspection shows that fake samples tend to be shorter than real samples across training, validation, and test splits. In the test set, fake samples have an average duration of 1.65 seconds, while real samples have an average duration of 2.52 seconds. In the training set, fake samples average 1.67 seconds, while real samples average 3.45 seconds.

C. Feature Extraction

wav2vec 2.0 BASE is used as a frozen self-supervised speech feature extractor. No task-specific fine-tuning is applied; instead, ablation study on the wav2vec 2.0 BASE model was conducted to seek layers that give the best result. The ablation will take the bottom, middle and top layer of the wav2vec 2.0 base model. To be specific, it'll take layers with index 3, 7, and 11. The frozen feature will be used to train an MLP classifier to retrieve its evaluation information therefore determining which layer performs the best. After the initial ablation study, the layer proved to be of robust performance is the Layer 4 (index 3). Therefore, hidden states are extracted from wav2vec 2.0 layer 4 (index 3). For each segment, temporal mean pooling is applied across time steps to produce a single 768-dimensional vector. The resulting embedding vector is used as input to the downstream classifiers.

D. Class Imbalance Handling

Segmentation caused imbalance in the dataset because of skipped clips. Inspection on the training set shows imbalance, with fake samples representing 32.0% of the training data. To reduce bias toward the majority class, SMOTE is applied to the training set only. Synthetic minority-class examples are generated in the 768-dimensional embedding space. Validation and test sets are not oversampled, ensuring that evaluation reflects the original data distribution. SMOTE uses the k-NN algorithm to synthesize the k nearest neighbors centered on the random samples of the minority class. The making of new synthetic samples is done in two steps. First is selecting samples from a random minority class and second is applying the k-NN algorithm to choose the location to generate synthetic samples.

If the number of missing samples is greater than the number of samples available in the minority class, then all samples in the minority class are selected. If the other opposite occurred, then instead randomly select the subset of all samples. After the selection, the k nearest neighbors of the minority class random samples that have a linear relationship with the selected minority class samples are selected.

E. MLP Classifier

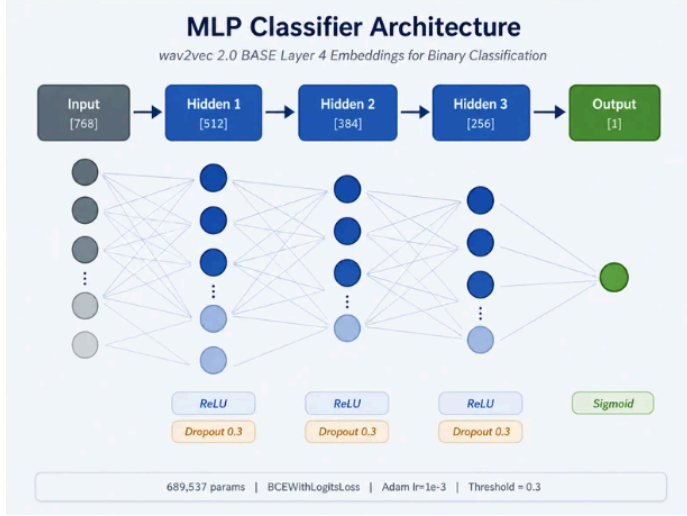


Fig. 2. Architecture of the selected MLP classifier using 768-dimensional wav2vec 2.0 embeddings as input.

The selected MLP architecture consists of three hidden layers with dimensions [512, 384, 256]. ReLU activation is used in hidden layers, and dropout with probability 0.3 is applied to each hidden layer. The final output layer produces one logit for binary classification. Sigmoid activation is applied only during inference to obtain the fake-class probability. Training uses Adam optimization with learning rate 1×10^{-3} and BCEWithLogitsLoss. Early stopping with patience 5 monitors validation loss. The selected MLP contains 689,537 trainable parameters.

F. Inference Threshold Calibration

Inference threshold (often called a decision threshold or classification threshold) is the specific cutoff point used to convert a model's continuous probability score into a definitive decision. Because the training and test class ratios differ, the model might have a bias towards certain labels. A fixed threshold of 0.5 may not provide the best operating point. Therefore, thresholds of 0.3, 0.4, 0.5, and 0.6 are evaluated. The calibration focuses on the balance of the model itself by F1-score while considering precision and recall trade-offs. In this study, threshold 0.3 provides the highest F1-score and is selected as the final operating threshold.

G. Explainable AI with SHAP GradientExplainer

SHAP GradientExplainer is applied to the trained MLP to estimate the contribution of each wav2vec embedding dimension to the output prediction. Global feature importance is computed as the mean absolute SHAP value for each of the 768 dimensions. The resulting ranking identifies the most influential dimensions and supports cumulative feature importance analysis. Since wav2vec dimensions are latent representations, the interpretation is limited to model-level feature contribution rather than direct physical acoustic meaning.

H. Cumulative SHAP-Based Feature Subset Analysis

Embedding dimensions are ranked according to mean absolute SHAP value. Cumulative importance is then computed from the ranked list. This analysis identifies how many dimensions are required to retain different proportions of total attribution mass. In the final output, K=200 retains 55.1% of cumulative importance, K=374 retains 80.0%, and K=572 retains 95.0%. These results are used only as feature importance analysis; retraining a reduced-input MLP using the selected subsets is left for future work.

I. Evaluation Metrics

Model performance is evaluated using accuracy, precision, recall, F1-score, and AUC. Confusion matrices are used to inspect false positives and false negatives. Precision is especially important in this task because falsely accusing authentic speech as fake can create reputational or forensic risk. Recall is also important because low recall means fake samples may pass as real.

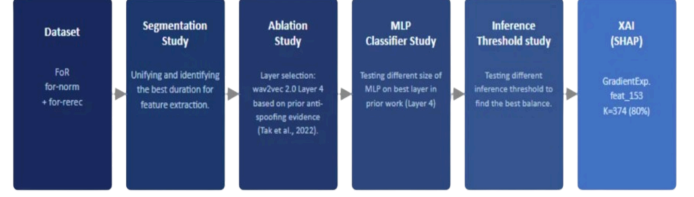


Fig. 3. Proposed audio deepfake detection pipeline using wav2vec 2.0 embeddings, MLP classification, and SHAP explainability.

IV. RESULTS AND DISCUSSION

This section will serve several results in order: the duration results, the ablation result, MLPs comparison, and final model one sample result. The mentioned evaluation matrices will be included as an evaluation tool.

A. Duration Study

Table 1 presents the inference test result on different duration configuration, using the wav2vec 2.0 layer 12 and inference threshold 0.5. The best performance lies on an upper threshold of 4 and lower threshold of 1.33. Compared to the model with an upper threshold of 4 and lower threshold of 2, the former yielded better performance by 0.0549 points. While the second best performing model uses upper threshold of 2 and lower threshold of 1. This suggests that for audio deepfake classification, length is needed to help the model seek patterns but the density of the data matters as well. A smaller lower threshold with the same upper threshold performs better than a bigger lower threshold. This seems to suggest that the 0 padding doesn't affect the performance of the model in a significant way.

B. Wav2vec 2.0 Ablation Study

Table II presents the inference test result on different frozen features on 3 layered MLP to represent ablation on the wav2vec 2.0 base model. The ablation is done on the bottom layer which is index 3 (layer 4), middle layer which is index 7 (layer 8) and top layer which is index 11 (layer 12). The inference uses 0.5 threshold.

The ablation results shows that contrary to conventional belief that deeper layers provide better representations, this ablation analysis reveals that early wav2vec 2.0 layers (layer 4) achieve 0.9336 accuracy, outperforming final layers by 0.1474 points.

Wav2vec 2.0 is designed for extracting speech features. The contextual encoding allows the model to learn the structure of speech; it learns to distinguish speech patterns from non-speech patterns (like background noise) in its latent representations. Therefore, higher, more complex layers may remove low level artifacts.

However the lower layers, which contain low level artifacts, consistently show higher accuracies than the higher layers as seen in figure 10 which contains higher level features with less artifacts. And from table 1, the lower layers show better performance overall.

This finding suggests that hoax audio detection relies instead more on low-level acoustic artifacts such as how words are said than high-level semantic content or what were the actual words being said in the audio.

C. MLP Architecture Variation

Table III summarizes MLP architecture variation results evaluated on the for-norm test set at threshold 0.6. The 3-layer [384, 192, 96] model achieves F1=0.9126, outperforming the 2-layer [256, 128] model with F1=0.9076. Increasing depth does not consistently improve performance. The 4-layer [512, 384, 256, 128] and 5-layer [512, 384, 256, 128, 64] models achieve lower recall and lower F1-scores, suggesting that larger depth may increase conservativeness or overfitting under the evaluated threshold. The 4-layer [1024, 512, 256, 128] model obtains the highest F1 in this specific ablation setting, but it has substantially more parameters. Therefore, the final model prioritizes the optimized 3-layer [512, 384, 256] architecture because it balances capacity, precision, and explainability with fewer parameters than the largest model.

B. Threshold Calibration

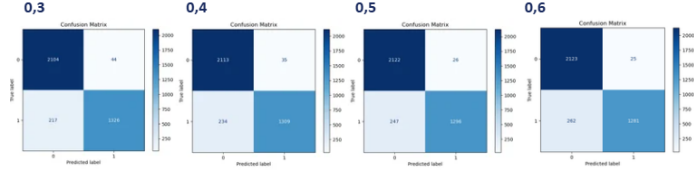


Fig. 4. Threshold calibration results for the selected MLP classifier.

Table IV shows the threshold calibration results for the selected MLP [512, 384, 256]. Precision remains consistently high across all thresholds, staying above 98%. Lowering the threshold from 0.6 to 0.3 increases recall from 83.08% to 85.55% and improves F1-score from 90.19% to 91.41%. The precision decrease is minimal, from 98.62% to 98.14%. Therefore, threshold 0.3 is selected as the optimal operating point.

C. Classifier Comparison

Table V compares the selected MLP with Random Forest and XGBoost classifiers trained on the same wav2vec feature representation. XGBoost achieves the highest raw performance, with 94.83% accuracy and 93.65% F1-score. However, the MLP achieves the highest precision at 98.14%, indicating that it produces fewer false alarms against real speech. This is important because false positives in deepfake detection may incorrectly label authentic speech as fake. The MLP is also compatible with gradient-based SHAP explanation, while XGBoost does not natively support SHAP GradientExplainer in the same differentiable neural setting. Therefore, the MLP is selected as the final model despite XGBoost achieving the highest raw F1-score.

TABLE I
DURATION VARIATION STUDY

Upper Threshold (s)	Lower Threshold	Acc.	Prec.	Rec.	F1
2	0.6	0.7151	0.8483	0.4320	0.5725
4	1.33	0.7862	0.9333	0.5262	0.6730
6	2	0.8582	0.8826	0.4835	0.6247
2	1	0.7608	0.9665	0.4703	0.6328
4	2	0.8575	0.8810	0.4761	0.6181
6	3	0.8996	0.8919	0.3367	0.4889

TABLE II
LAYERWISE ABLATION STUDY ON WAV2VEC 2.0

Layer	Acc.	Prec.	Rec.	F1
4	0.9336	0.9848	0.8548	0.9150
8	0.8662	0.9816	0.6928	0.8123
12	0.7862	0.9333	0.5262	0.6730

TABLE III
MLP ARCHITECTURE VARIATION RESULTS AT THRESHOLD 0.6 ON THE FOR-NORM TEST SET

Architecture	Params	Val F1	Acc.	Prec.	Rec.	F1
2-Layer [256, 128]	229,889	0.9987	0.9279	0.9783	0.8464	0.9076
3-Layer [384, 192, 96]	387,841	0.9978	0.9320	0.9857	0.8496	0.9126
4-Layer [512, 384, 256, 128]	722,305	0.9978	0.9090	0.9910	0.7894	0.8788
5-Layer [512, 384, 256, 128, 64]	730,497	0.9981	0.8981	0.9932	0.7615	0.8621
4-Layer [1024, 512, 256, 128]	1,476,609	0.9972	0.9317	0.9743	0.8594	0.9132

TABLE IV
THRESHOLD CALIBRATION RESULTS FOR MLP [512, 384, 256]

Threshold	Accuracy	Precision	Recall	F1
0.3 (optimal)	93.28%	98.14%	85.55%	91.41%
0.4	93.20%	98.64%	84.90%	91.26%
0.5	92.85%	98.63%	84.06%	90.76%
0.6	92.44%	98.62%	83.08%	90.19%

TABLE V
CLASSIFIER COMPARISON ON THE POOLED FOR TEST SET

Classifier	Acc.	Prec.	Rec.	F1	AUC
Random Forest	86.89%	80.87%	89.89%	85.14%	0.9408
XGBoost	94.83%	96.17%	91.25%	93.65%	0.9874
MLP [512, 384, 256]	93.28%	98.14%	85.55%	91.41%	0.9830

D. Confusion Matrix Analysis

At threshold 0.3, the final MLP produces $TN=2123$, $FP=25$, $FN=223$, and $TP=1320$. The false positive count is small compared with the number of real samples, supporting the claim that the model is conservative in flagging speech as fake.

However, the false negative count remains non-trivial, meaning some fake samples are predicted as real. This result shows that the selected threshold improves recall relative to higher thresholds but does not fully eliminate missed fake detections.

E. SHAP Global Feature Importance

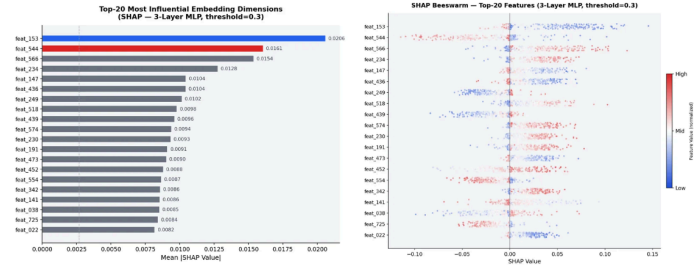


Fig. 5. SHAP global feature importance and beeswarm visualization for the selected MLP model.

SHAP GradientExplainer identifies feat 153 as the strongest discriminative embedding dimension. The top feature ranking is led by feat 153, followed by feat 544, feat 566, feat 234, feat 147, feat 436, feat 249, feat 518, feat 439, and feat 574. The SHAP beeswarm plot shows that high activation values for feat 153 correlate with fake predictions. This suggests that the trained MLP relies on a concentrated subset of wav2vec Layer 4 dimensions rather than using all dimensions equally.

This interpretation should be treated carefully. The dimensions of wav2vec embeddings are latent model representations, not direct acoustic descriptors such as pitch, energy, or formant frequency. Therefore, the SHAP result is best interpreted as evidence of model-level discriminative dimensions, not as a direct explanation of physical speech properties.

F. Cumulative SHAP Importance and Feature Subset Analysis

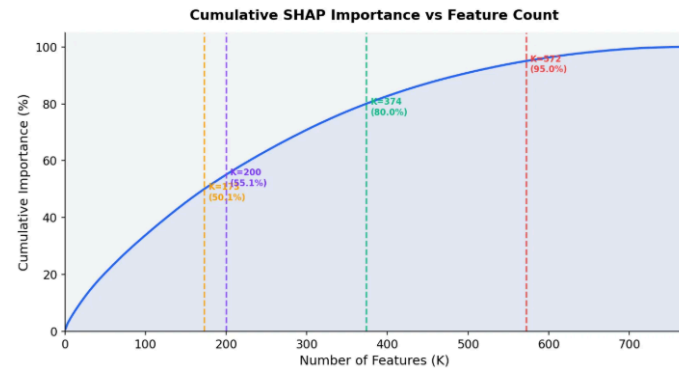


Fig. 6. Cumulative SHAP importance curve showing the number of embedding dimensions required to retain different levels of attribution importance.

The cumulative SHAP curve indicates that discriminative information is unevenly distributed across embedding dimensions. $K=200$ features retain 55.1% of cumulative SHAP importance, making this a compact subset. $K=374$ features are required to retain 80.0% of cumulative importance, while $K=572$ features retain 95.0%. These results correct the earlier interpretation that $K=200$ retained 80% importance. In the updated analysis, $K=374$ is the appropriate 80% threshold.

The practical implication is that a reduced-feature model may be possible, but it has not yet been empirically verified. Therefore, the present study reports SHAP-based feature subset analysis only. A retrained MLP using $K=200$ or $K=374$ selected dimensions should be evaluated in future work before making deployment or efficiency claims.

G. Preliminary Cross-Condition Analysis

The pooled FoR test setting includes both clean and channel-degraded conditions. The model achieves strong overall pooled performance, with $F1=91.41\%$ and $AUC=0.9830$. However, error inspection suggests that channel-degraded audio may increase false-negative risk. This is plausible because loudspeaker-microphone re-recording can alter frequency response, introduce room acoustics, and mask synthesis artifacts that may be useful for classification.

Importantly, this study does not yet report a complete separate metric breakdown for the for-rerec subset. Therefore, the correct conclusion is not that the model has definitively failed or succeeded under for-rerec, but that preliminary pooled-test inspection indicates a potential robustness issue. Full for-rerec subset evaluation, channel-aware training, and domain adaptation are left as future work.

H. Discussion

The results show four important findings. First, high validation performance does not guarantee equally high test performance, especially when the test set has a different fake-class ratio and includes more challenging acoustic conditions. Second, threshold calibration is useful because the final threshold 0.3 improves recall and F1-score while preserving very high precision. Third, explainability adds value beyond classification metrics by showing which embedding dimensions dominate the model decision. Fourth, The ablation study reveals that low-level acoustic artifacts preserved in early representations are more discriminative for deepfake audio detection than high-level semantic features.

The comparison with XGBoost also clarifies the trade-off between raw performance and research objective. XGBoost is the strongest raw classifier in this experiment, but the selected MLP better supports the explainability objective and produces the highest precision. For applications where false positives are costly, this conservative behavior can be desirable. However, for applications where catching as many fake samples as possible is more important, XGBoost may be preferred because of its higher recall and F1-score.

I. Limitations

This study has several limitations. First, evaluation is limited to the FoR dataset and has not yet been extended to external benchmarks such as ASVspoof, ADD, or in-the-wild datasets. Second, separate for-rerec subset metrics are not yet available, so the cross-condition result must be treated as preliminary. Third, SHAP explains the trained MLP but does not provide direct acoustic meaning for each wav2vec dimension. Fourth, the SHAP-based feature subset analysis has not yet been validated by retraining a reduced-input classifier. Fifth, the selected model uses frozen embeddings rather than end-to-end fine-tuning, which may limit adaptation to new domains. Lastly, hyperparameters such as threshold inference are chosen and evaluated by systematic manual trial. Automatic tuning is suggested for future development.

V. CONCLUSION

This paper presented an explainable audio deepfake detection framework using wav2vec 2.0 Layer 4 embeddings and an MLP classifier. The final MLP [512, 384, 256] with threshold 0.3 achieved 93.28% accuracy, 98.14% precision, 85.55% recall, 91.41% F1-score, and 0.9830 AUC on the pooled FoR test set. XGBoost achieved the best raw F1-score at 93.65%, but the MLP was selected because it provided the highest precision and supported SHAP GradientExplainer.

The layer-wise ablation analysis reveals a counter-intuitive yet significant finding: early transformer layers (layer 4) of the pre-trained wav2vec 2.0 model outperform deeper layers by 0.1474 points (0.9336 vs 0.7862 accuracy), demonstrating that low-level acoustic artifacts preserved in early representations are more useful for deepfake audio detection than high-level semantic features. This challenges conventional assumptions about feature abstraction in pre-trained speech models and provides valuable guidance for researchers in this domain and opens the potential for a simpler yet potentially more efficient audio deep hoax detection system architecture.

SHAP analysis identified feat 153 as the most influential embedding dimension. Cumulative SHAP analysis showed that K=200 features retain 55.1% of total importance, K=374 features retain 80.0%, and K=572 features retain 95.0%. These values correct the previous interpretation of the feature subset results. Preliminary pooled-test error inspection suggests possible false-negative risk under channel-degraded audio, but full separate for-rerec evaluation remains future work.

Future work will focus on complete for-rerec subset evaluation, cross-dataset testing on recent benchmarks, channel-aware domain adaptation, multi-layer wav2vec feature fusion, reduced-feature MLP retraining using SHAP-selected dimensions, and lightweight deployment analysis.

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